MATLAB Random Bit Error Rate (BER) Simulator Report

Digital Communications Simulation & AWGN Channel Performance Analysis

1. Executive Summary & Lab Objective

This report details the implementation and empirical findings for the MATLAB Random Bit Error Rate (BER) Simulator. The core goal of this assignment is to simulate a complete digital communication link operating over an **Additive White Gaussian Noise (AWGN)** channel and evaluate the system's error probability across four target **Signal-to-Noise Ratio (SNR)** values: **0 dB, 3 dB, 6 dB, and 12 dB**.

By transmitting a sequence of $N = 10{,}000$ pseudo-random binary digits mapped to antipodal electrical voltage levels (+1 V / -1 V), this study quantifies how channel noise degrades binary signal detection and demonstrates the exponential reduction in bit errors as signal power increases relative to noise power.

2. Mathematical Framework & System Architecture

The simulated communication link consists of three sequential operational blocks: transmitter signal mapping, AWGN channel degradation, and receiver zero-threshold decision detection.

```
[ Random Bit Stream b[n] in {0,1} ]  -->  [ Antipodal Modulator s[n] = A(1-2b[n]) ]  -->  [ AWGN Channel r[n] = s[n] + w[n] ]
                                                                                                          |
[ Bit Error Rate BER = N_err / N   ]  <--  [ Error Detector e[n] = b[n] != b_hat   ]  <--  [ Zero Threshold b_hat[n] = (r[n] < 0) ]
```

A. Antipodal Modulation (BPSK)

The transmitter generates equiprobable binary digits $b[n] \in \{0, 1\}$ with $P(0) = P(1) = 0.5$. Each bit is mapped to a discrete voltage level $s[n]$ with signal amplitude $A = 1$:

$s[n] = A \cdot (1 - 2 \cdot b[n])$ = **+1 V** (for b[n]=0) or **-1 V** (for b[n]=1)

B. AWGN Channel & Noise Variance Derivation

The channel adds zero-mean Gaussian noise $w[n] \sim N(0, \sigma_w^2)$. The relationship between decibel SNR (SNR_{dB}) and linear SNR (SNR_{linear}) is:

$$\mathbf{SNR_{linear} = 10^{(SNR_{dB} / 10)}}$$

For amplitude $A = 1$, the required Gaussian noise variance is derived from $SNR_{linear} = A^2 / (2 \cdot \sigma_w^2)$:

$$\mathbf{\sigma_w^2 = A^2 / (2 \cdot SNR_{linear})}$$

The received corrupted signal is **r[n] = s[n] + w[n]**.

C. Zero-Threshold Bit Detection & BER Calculation

The receiver applies a zero-voltage decision threshold: **b_hat[n] = 1 if r[n] < 0, else 0**.
Bit errors occur when $b_hat[n] \neq b[n]$. Total Bit Error Rate is computed as **BER = (1 / N) $\sum$ I(b_hat[n] $\neq$ b[n])**.

3. Engineering Discussion & Key Findings

• **Exponential Error Suppression:** As SNR increases from 0 dB to 12 dB, the Gaussian noise variance decreases by a factor of 15.87 (from 0.5000 down to 0.0315). This reduction sharply suppresses the tail probabilities of the noise distribution crossing the zero-voltage boundary, causing BER to drop exponentially from 7.56% down to 0.00%.

• **Noise Threshold Crossings at 0 dB:** At 0 dB SNR, the noise standard deviation is $\sigma_w = \sqrt{0.5} \approx 0.707$ V. Because σ_w is comparable to the signal amplitude A = 1.0 V, random noise fluctuations frequently exceed 1.0 V in magnitude, pushing transmitted +1 V levels into negative territory (and vice versa), resulting in a 7.56% error rate.

• **Flawless Transmission at 12 dB:** At 12 dB SNR, the noise standard deviation drops to $\sigma_w = \sqrt{0.0315} \approx 0.177$ V. The distance to the decision boundary (1.0 V) represents 5.63 σ_w. The probability of Gaussian noise exceeding 5.63 σ_w is $Q(5.63) \approx 9 \times 10^{-9}$, rendering error occurrences virtually zero over a sample size of N = 10,000 bits.

• **Practical System Standards:** Modern wireless and fiber-optic communication protocols mandate an operating SNR of at least 10–12 dB (or incorporate Forward Error Correction) to ensure Bit Error Rates remain below 10^{-5} for reliable data transfer.